RoboColosseum

Evaluating Physical Intelligence at Scale in the Wild

RoboColosseum: Evaluating Physical Intelligence at Scale In-The-Wild

NUS • University of Washington • Stanford • UNC • Allen Institute for AI • Oxford • Hugging Face • BitRobot • Singapore Embodied AI Centre

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TL;DR: network first; one ranking can’ t capture physical intelligence

RoboColosseum evaluates 5 VLAs and 3 frontier agents on **a distributed real-robot network staffed by full-time operators**: **zero-shot agents match specialized VLAs**, but open-world success is low for all; rankings across embodiments and tracks **disagree**; on the G1 humanoid, **no policy completed any task**.

5,699
Blind Open Track tests pairwise comparisons

124
real evaluation sites (offices/homes/ factories…)

4
embodiments SO-101 / Franka / YAM / G1

5 + 3
VLA policies + frontier AI agents

20 × 25
fine-tune tasks × trials per pair

Evaluation period: 2026-09-17 to 2026-10-28 (41 days). **Fine-tune Track:** 100 on-site teleoperated demos per task; 25 evaluation trials.

Motivation

Real-robot dilemma: control for comparability, diversity for trust

Simulation benchmarks

James et al. 2020, Li et al. 2025, etc.

- Repeatable resets, controlled variation
- Gaps in perception, calibration, dynamics, appearance
- → Poor predictor of real-robot performance

Single-site real robot

FurnitureBench, RoboDojo, etc.

- Unified hardware and task protocols
- Resets, maintenance, execution all manual
- → Few rollouts, high-variance estimates

Distributed A/B evals

RB2 、 RoboArena

- RoboArena: 7 universities, 612 comparisons
- Single embodiment only (DROID/Franka)
- → Volunteer-driven, hard to scale sustainably

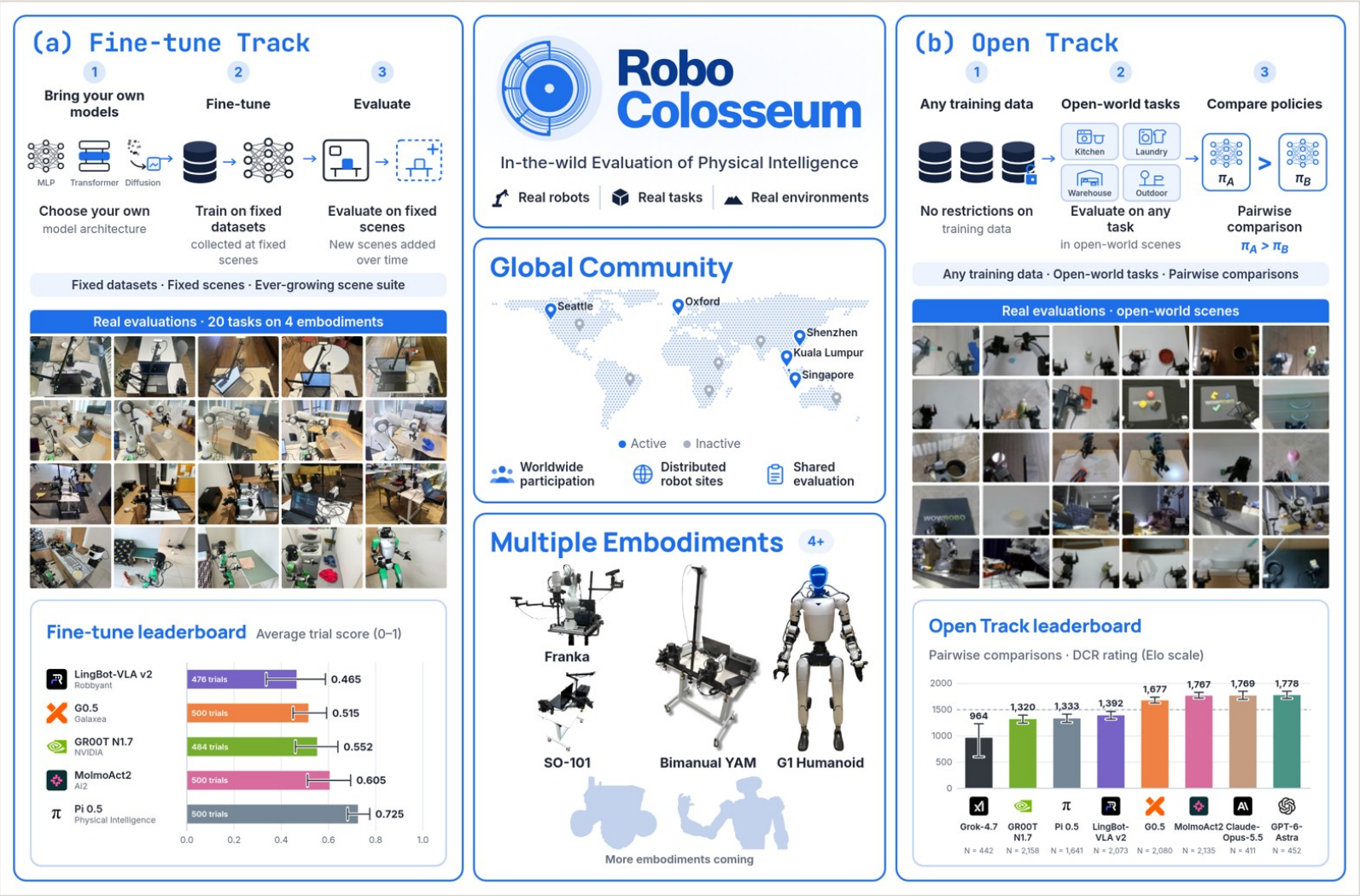
RoboColosseum's answer

Up to 30 full-time operators • 100+ robots (US / UK / Singapore / Malaysia / China)

Unified hardware + unified policy serving + two tracks, each stating what is fixed vs. free

Method

Framework: distributed sites, unified serving, two complementary tracks



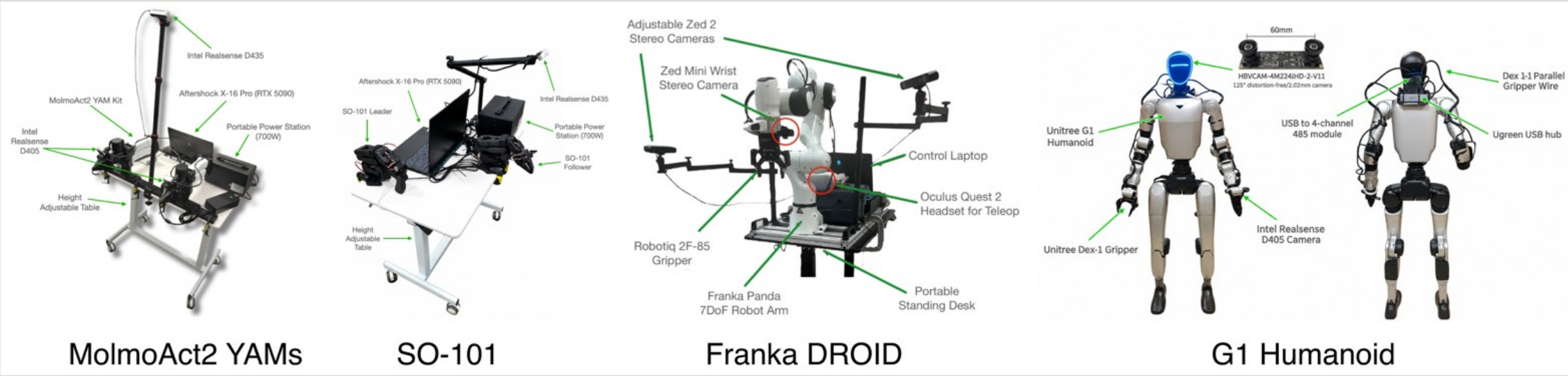
Fine-tune Track

Fixed 100 demos + fixed task cards; compares learning a new task from the same data.

Open Track

No task adaptation; tasks and scenes vary freely, but conditions within each A/B comparison are identical.

4 embodiments: low-cost & research arms, bimanual, humanoid



Bimanual-YAM

14-DoF joints (6+1 per arm)

1 overhead + 2 wrist cameras

GELLO teleoperation

SO-101

6-DoF joints (5+1)

1 external + 1 wrist camera

Leader–follower teleoperation

Franka (DROID)

8-DoF joints (7+1)

2 external + 1 wrist camera

DROID Quest teleoperation

Unitree G1

Walking humanoid + parallel gripper

Fine-tune Track only

Pico VR + whole-body controller

Method

Both tracks state explicitly what is fixed and what is free

	Fine-tune Track	Open Track
Measures	Adapting to new tasks from fixed demos	Zero-shot generalization, no task adaptation
Fixed	100 on-site teleop demos per task (public) Unified task cards, success & progress criteria	Within each A/B comparison: same robot, instruction, objects, initial layout and timeout
Free	Each model’ s official recipe (hyperparams, steps differ)	Tasks and environments vary across comparisons
Scale	4 embodiments × 5 tasks = 20 tasks; 25 trials per pair	5,699 comparisons, 124 sites, 3 embodiments
Metrics	Success rate SR + progress score PS = mean(s_n / K)	DCR rating (Elo scale), blind win/tie/loss
Entrants	5 VLAs (fine-tuned from base checkpoints)	5 VLAs + 3 frontier agents

Open Track ranking jointly estimates ability and task difficulty

1

Blind assignment

System assigns two anonymous models, A and B

2

Identical conditions

Run in turn on the same robot and task setup

3

Record

W/T/L, per-model progress, written feedback, difficulty 1–10

4

Joint inference

Ability θ, difficulty β and difficulty-prior parameters fit jointly

Success probability (Rasch / 1PL)

$$P(y=1) = \sigma(\theta_p - \beta_t)$$

θ_p: policy ability; β_t: task difficulty

Human-rated difficulty as prior

$$\beta_t \mid d_t = k \sim N(\mu_k, \tau^2),\ \mu_1 < \ldots < \mu_{10}$$

Physical outcomes can pull a task away from its category mean

Leaderboard (Elo scale, linear transform)

$$R_p = 1500 + (400 / \ln 10) \cdot E[\theta_p \mid D]$$

Order-preserving; not match-by-match Elo updates

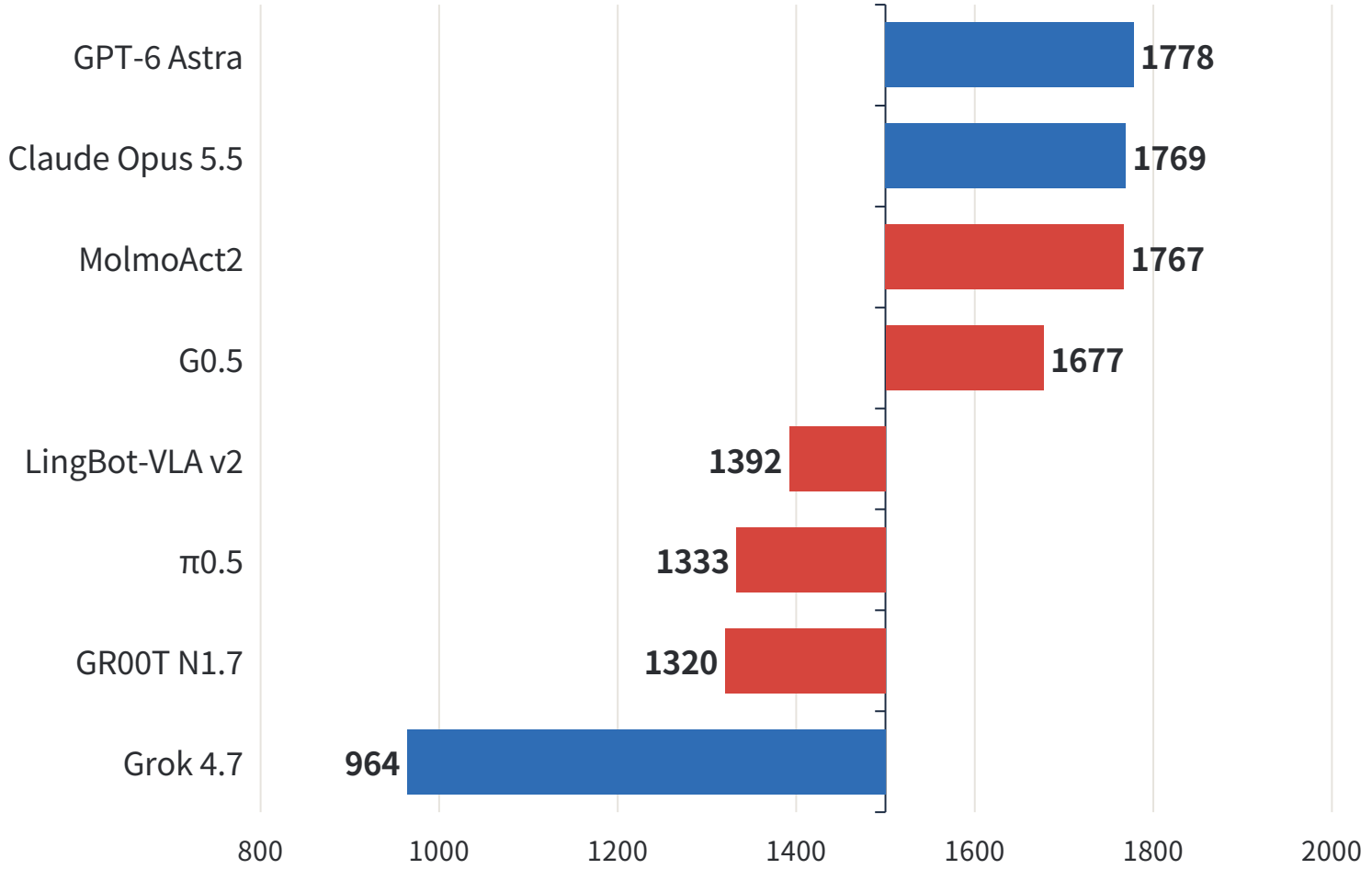
Active pairing (as in Chatbot Arena)

$$P_t(a) \propto \sqrt{(\Sigma_{aa} / n_a)} - \sqrt{(\Sigma_{aa} / (n_a+1))}$$

Prioritizes uncertain, under-sampled model pairs

Global ranking: frontier agents tie the best VLA at top; Grok last

Global DCR rating (Elo scale): bars extend left/right from the 1500 mean



1778 ≈ 1769 ≈ 1767

GPT-6 Astra / Claude / MolmoAct2 share the top tier

964

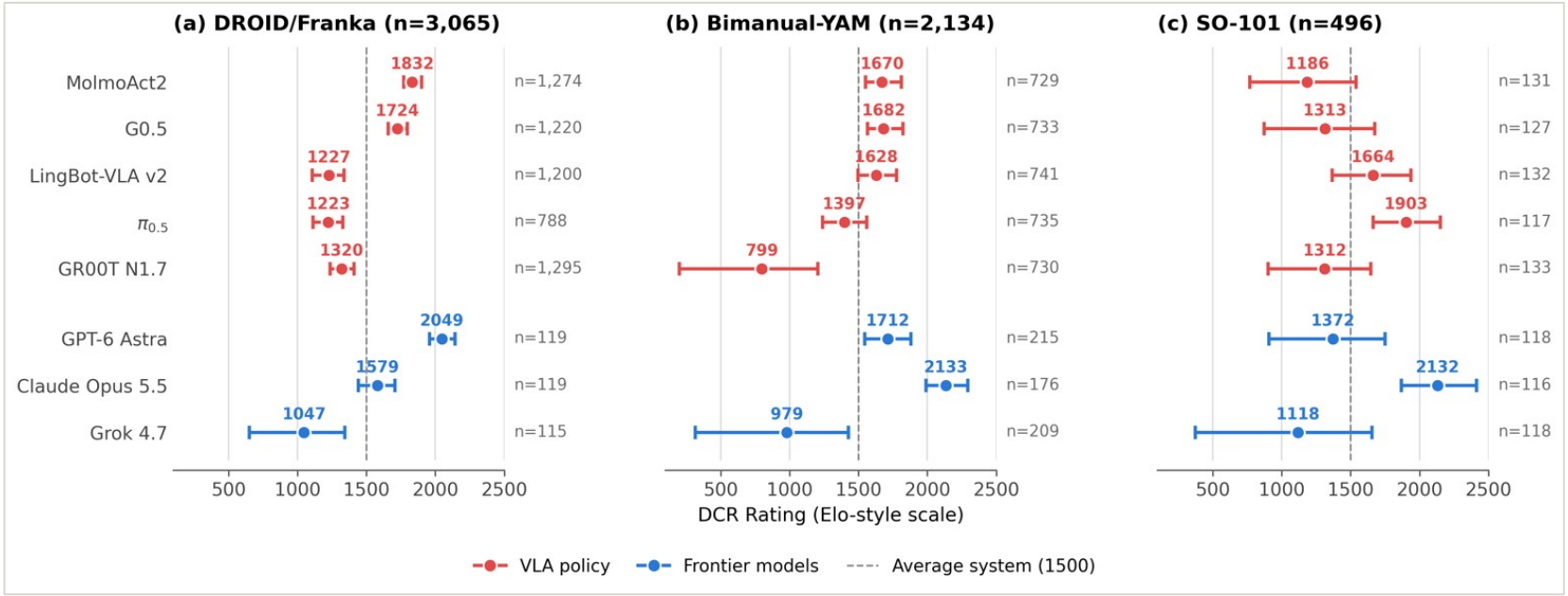
Grok 4.7 far below average: agents vary widely

No fine-tuning

Agents control robots via joint-position tool calls, no demos

Source: paper Figure 1 Open Track leaderboard (DCR, Elo scale)

Yet the leader varies: each embodiment has a different champion



DROID / Franka (n=3,065)

agent： GPT-6 Astra 2049
VLA： MolmoAct2 1832 • G0.5 1724

Bimanual-YAM (n=2,134)

agent： Claude Opus 5.5 2133
VLA: G0.5 1682 • GR00T only 799

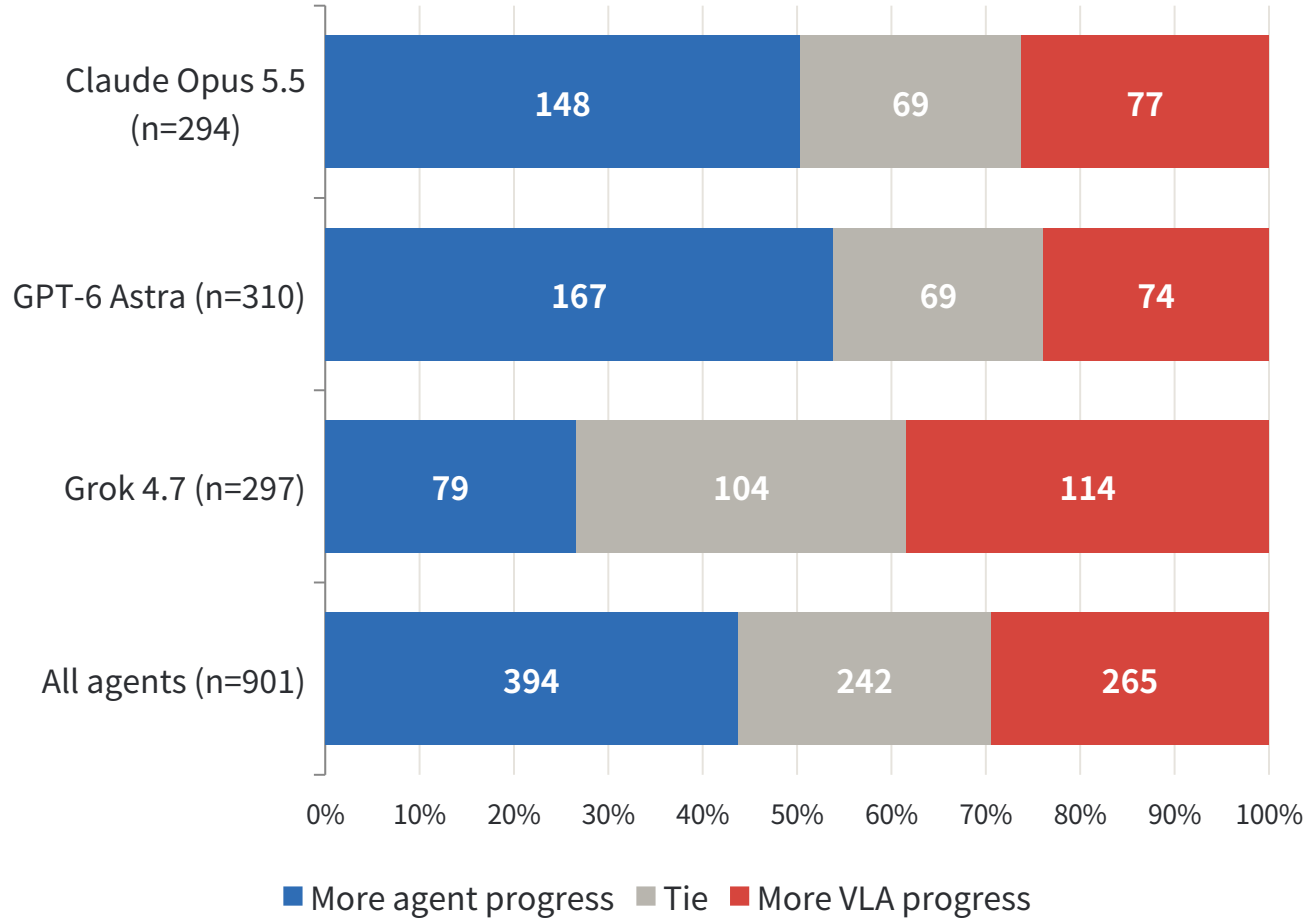
SO-101 (n=496)

agent： Claude Opus 5.5 2132
VLA: π0.5 1903; few samples, wide interval

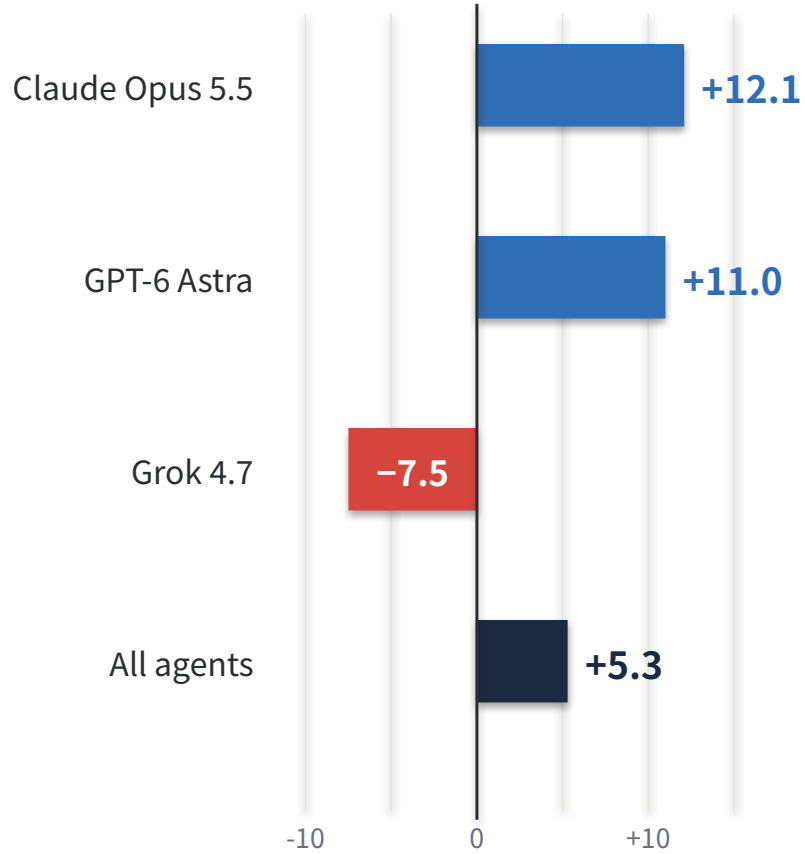
Source: paper Figure 3 (95% credible intervals; n = evaluations)

Same task: agents avg. +5.3 pp progress; the three vary widely

(a) Win / tie / loss counts (by progress)



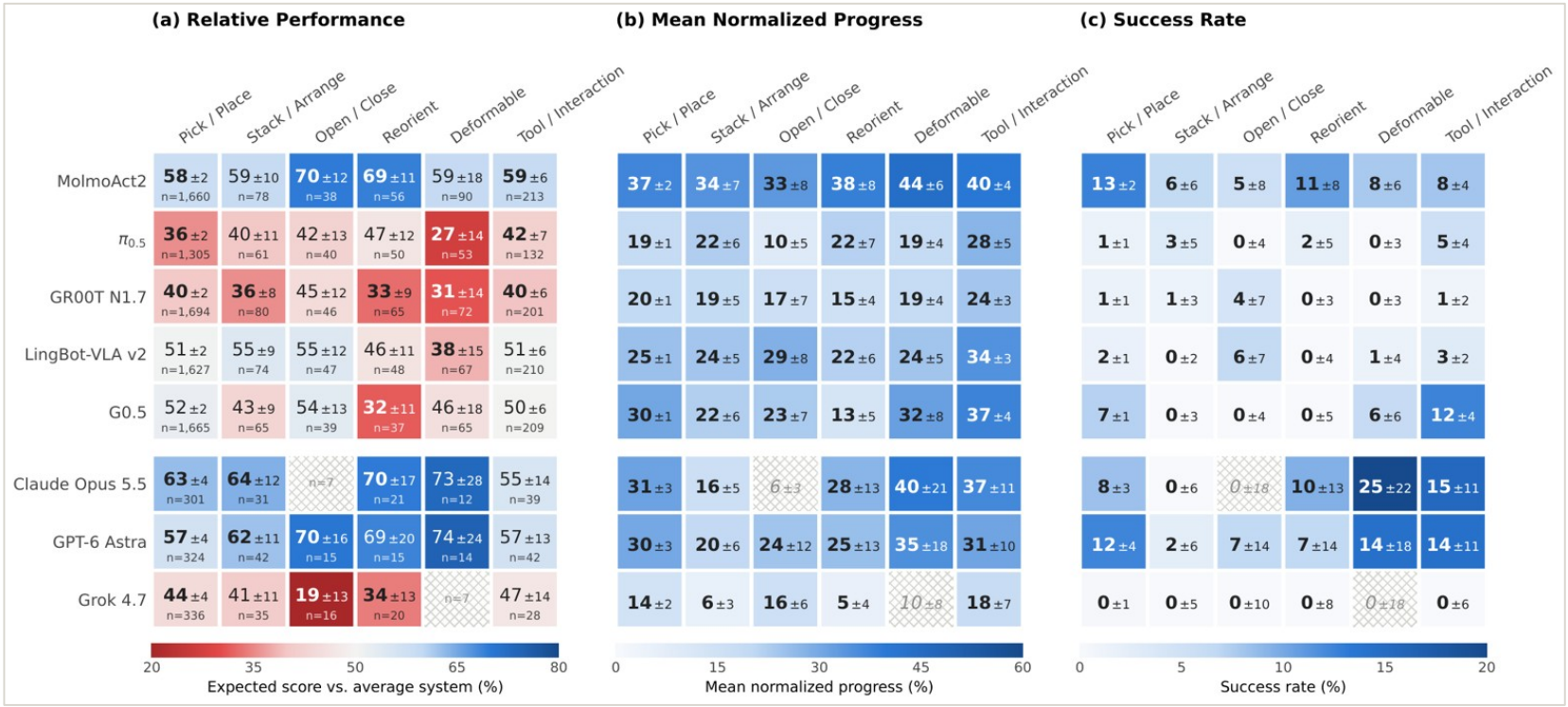
(b) Mean progress gap, agent − VLA (pp)



Positive = more agent progress; all-agent 95% CI excludes 0

Source: paper Figure 4 (901 agent–VLA pairs, normalized task progress)

Relative wins ≠ completion: open-world success rates are very low



58% – 77%

MolmoAct2 relative score: best VLA in all six categories

6% – 13%

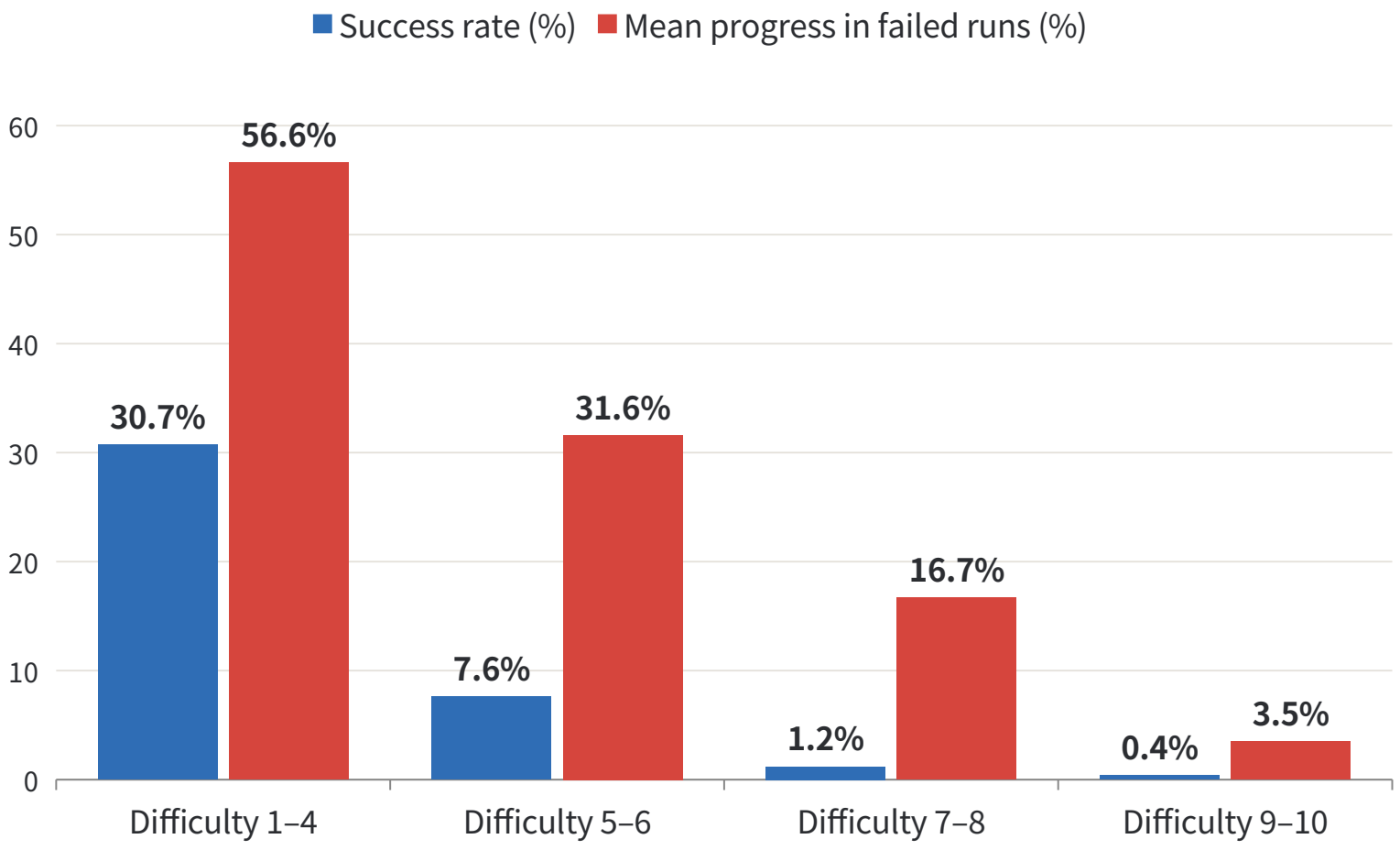
MolmoAct2's actual task success rate

⩽ 13%

Paper: no system exceeds this in any category (except Claude, deformable, 25%±22, n=12)

Source: Fig. 5 (6 categories; (a) progress BTD expected score (b) mean progress (c) SR)

Harder tasks: policies fail to even complete the first steps



Runs n: levels 1–4 374 • 5–6 5,376 • 7–8 4,248 • 9–10 1,394

30.7% → 0.4%

Success plunges with difficulty

56.6% → 3.5%

Mean progress in failures also nears zero

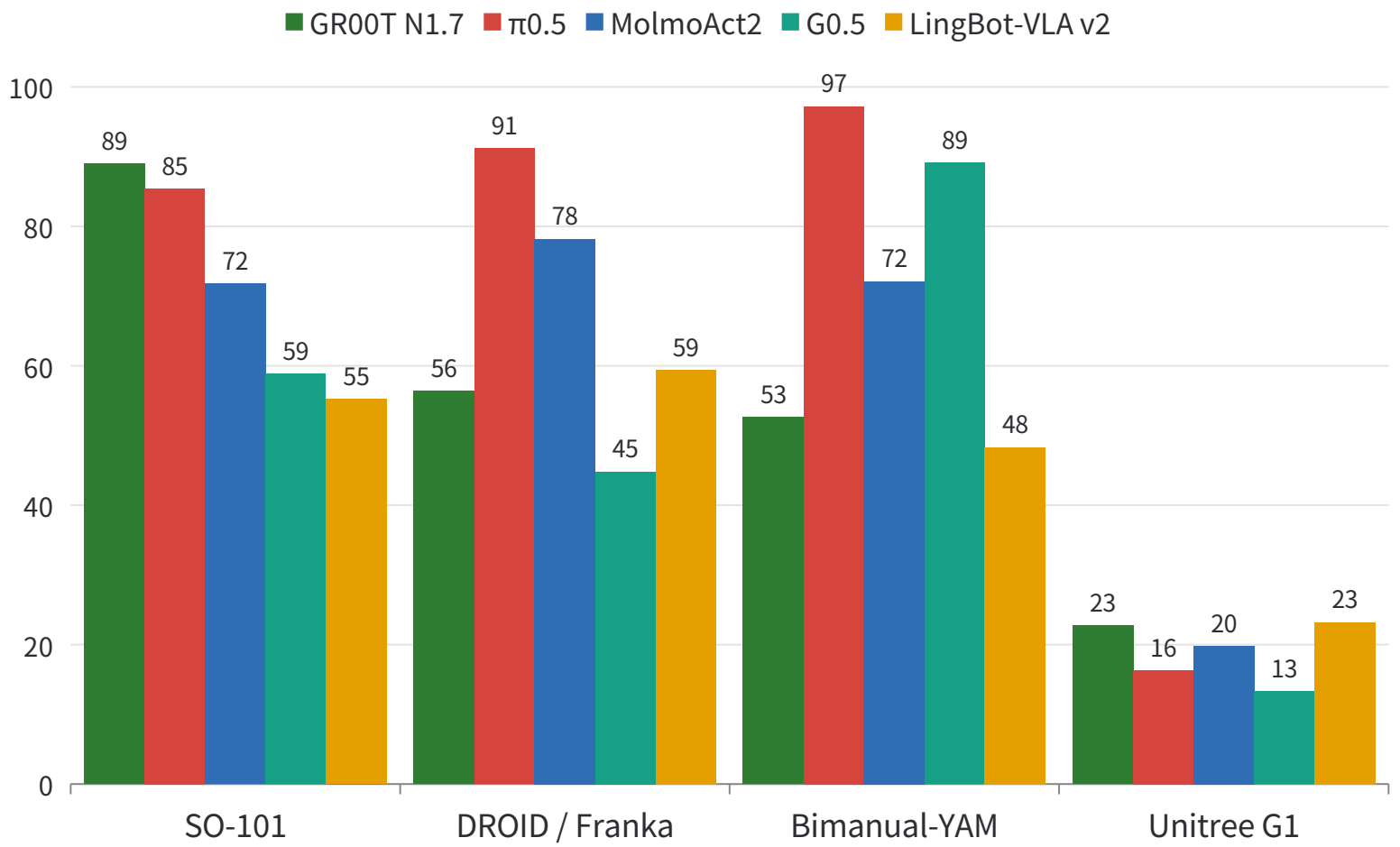
97%

Runs concentrate at difficulty 5–10 (11,018 / 11,392)

Source: paper Figure 6 (by human-rated difficulty; each A/B eval adds two runs)

Fine-tuned: π0.5 steadiest, GR00T tops SO-101, all fail on G1 humanoid

Mean progress score PS per embodiment (%, 5 tasks equally weighted)



Winner per embodiment (by SR)

SO-101

GR00T N1.7 SR 82.4%

DROID / Franka

π0.5 SR 84.0%

Bimanual-YAM

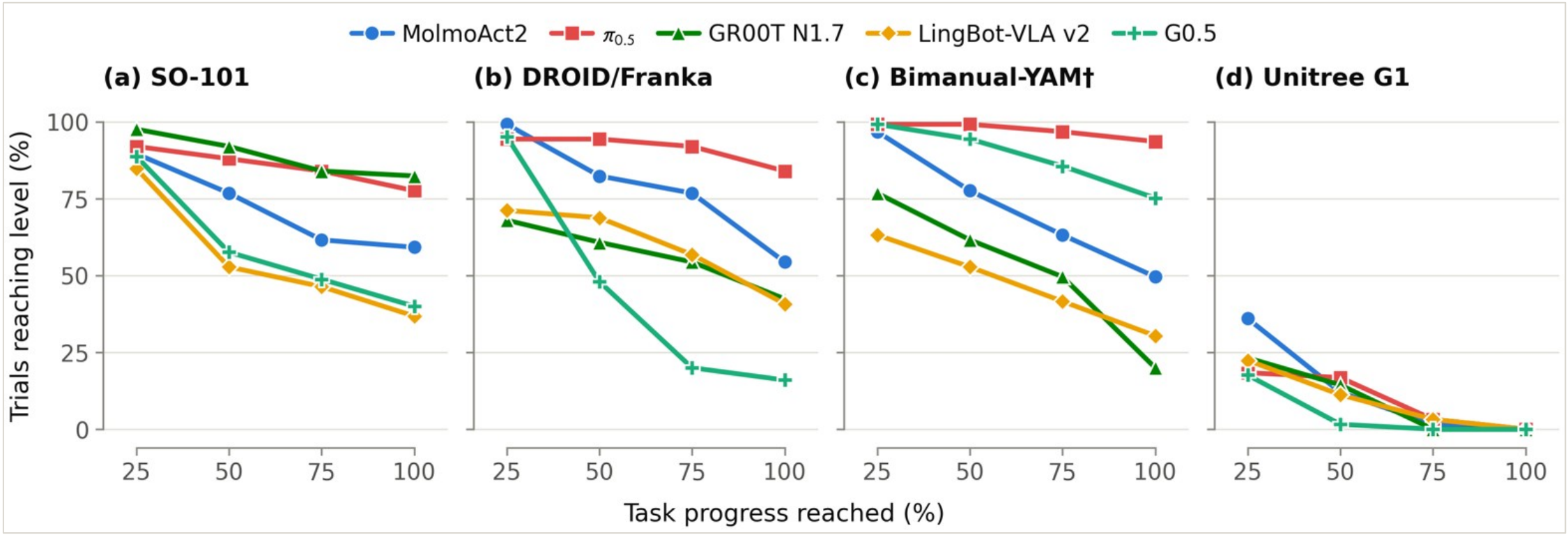
π0.5 SR 93.6%

Unitree G1

No success All SR 0% (PS 13.3–23.2)

Source: paper Table 2 (25 trials per model–task; 5 tasks equally weighted per embodiment)

Progress reveals what SR hides: “nearly done” vs. “never started”



~100% → 16%

G0.5 on DROID: almost always starts, rarely finishes

93.6%

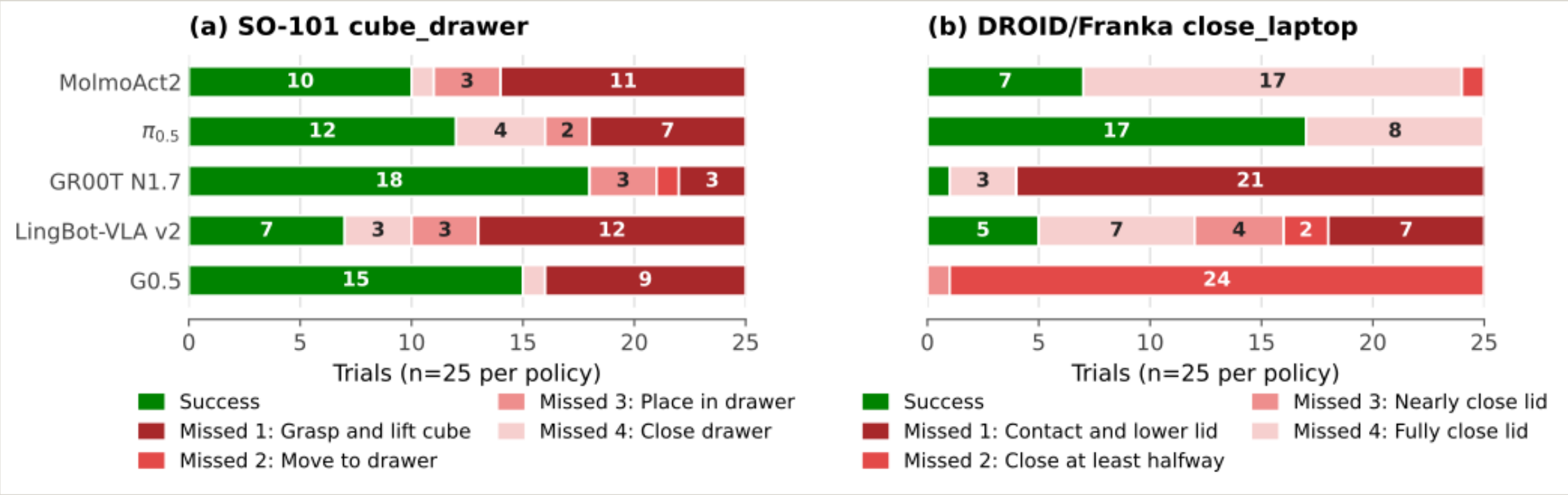
π0.5 on YAM keeps high coverage throughout and finishes

< 40%

G1: most trials fail to meet even 1/4 of criteria

Source: paper Figure 7 (share of trials reaching each progress threshold; 100% point = SR)

Same demos, different bottlenecks: models stall at different stages



SO-101 cube_drawer: shared difficulty at step one

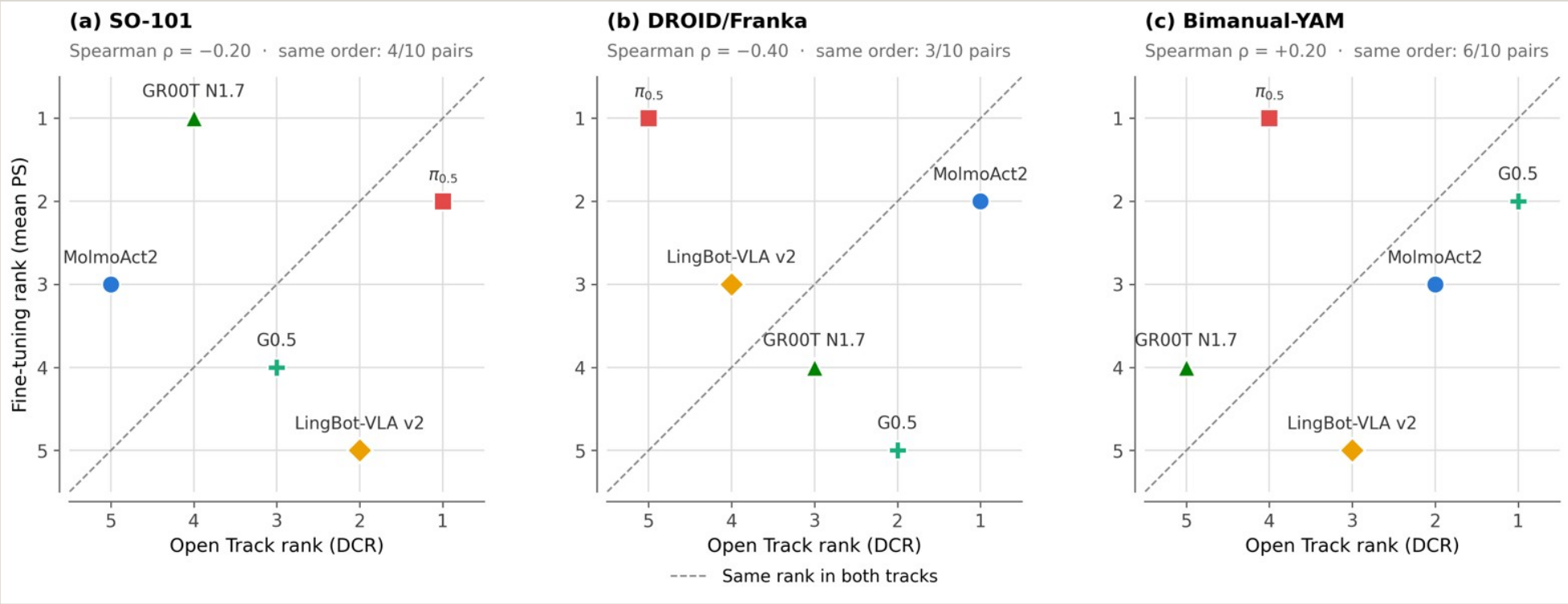
"Grasp and lift cube" unmet in 12%–48% of trials (GR00T 3 → LingBot 12)

DROID close_laptop: bottlenecks differ

G0.5: 24/25 not even half-closed • MolmoAct2 nearly closes but succeeds only 7× • π0.5 succeeds 17× • GR00T stuck at step one 21×

Source: paper Figure 8 (25 trials grouped by first unmet criterion)

Open Track rankings barely predict fine-tuned rankings



SO-101

ρ = −0.20

4 / 10 pairs agree in order

DROID / Franka

ρ = −0.40

3 / 10 pairs; π0.5 5th → 1st

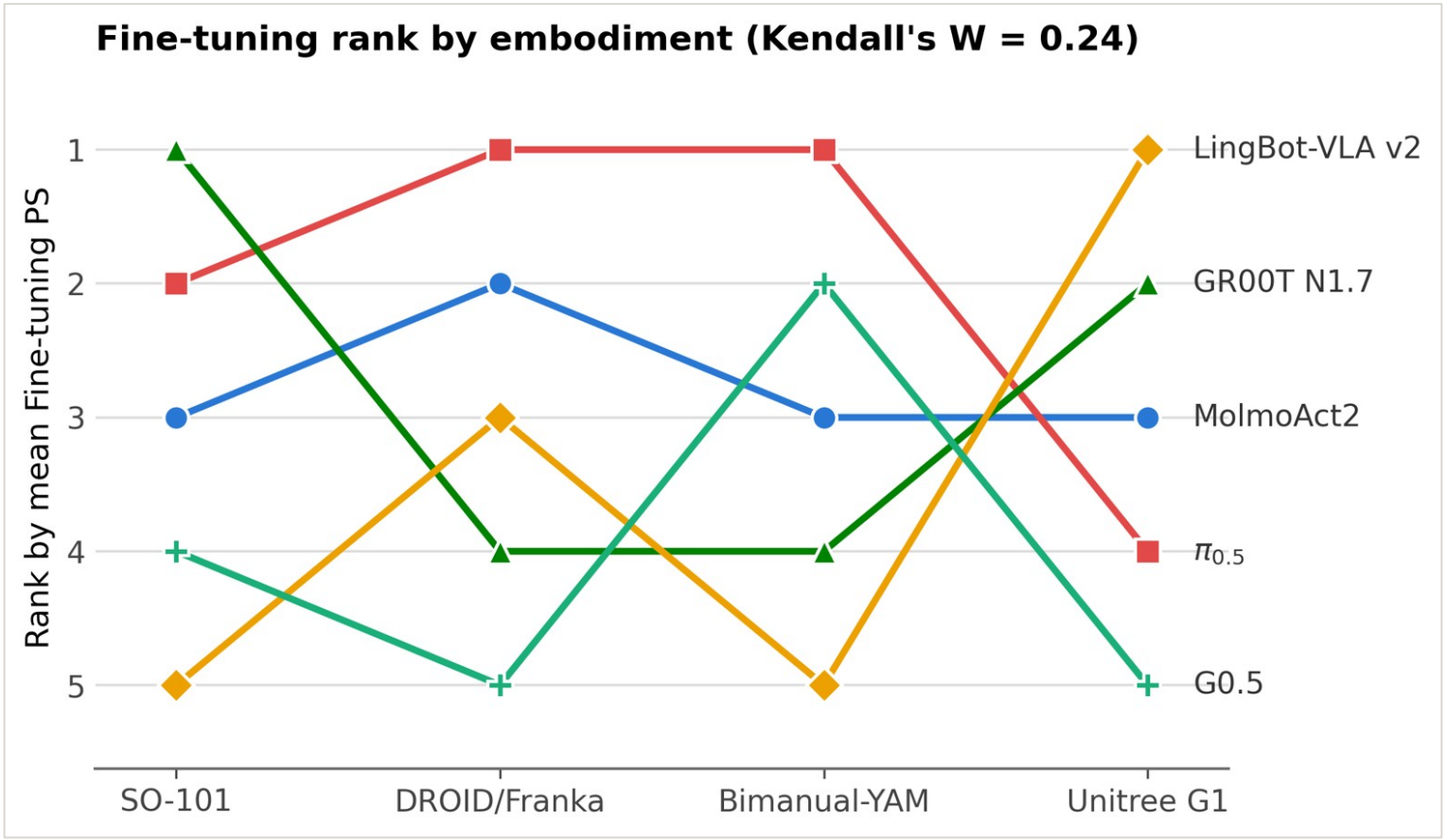
Bimanual-YAM

ρ = +0.20

6 / 10 pairs; π0.5 4th → 1st

Source: paper Figure 9 (Open: DCR rank; Fine-tune: mean PS rank)

Rankings also unstable across embodiments: Kendall's W only 0.24



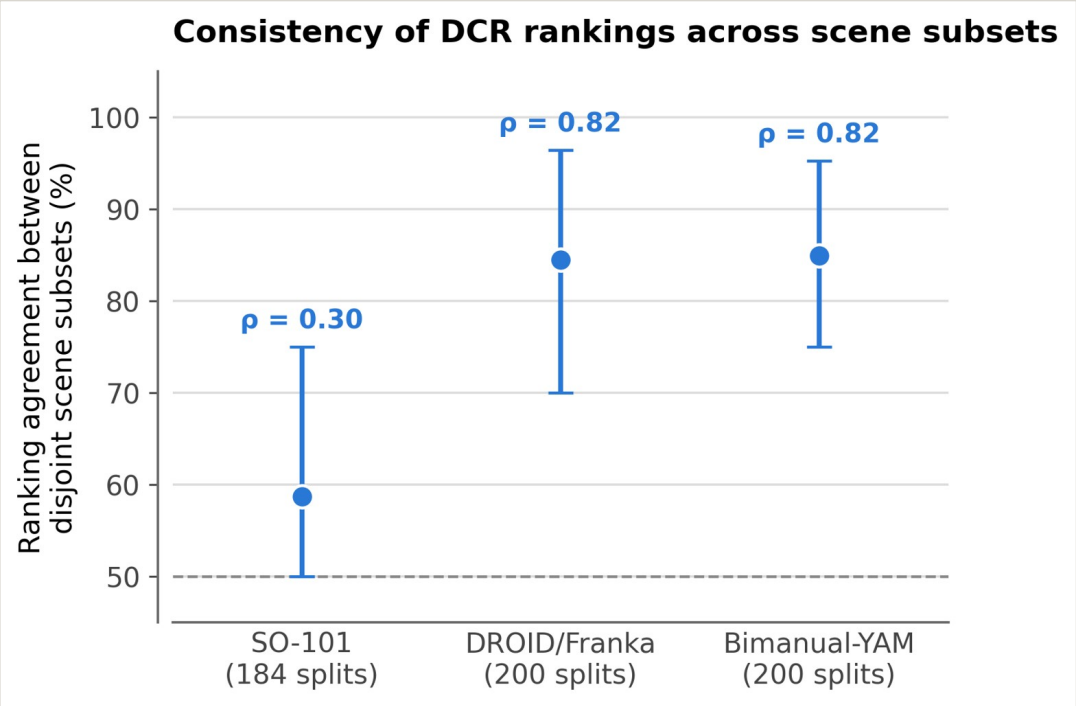
W = 0.24

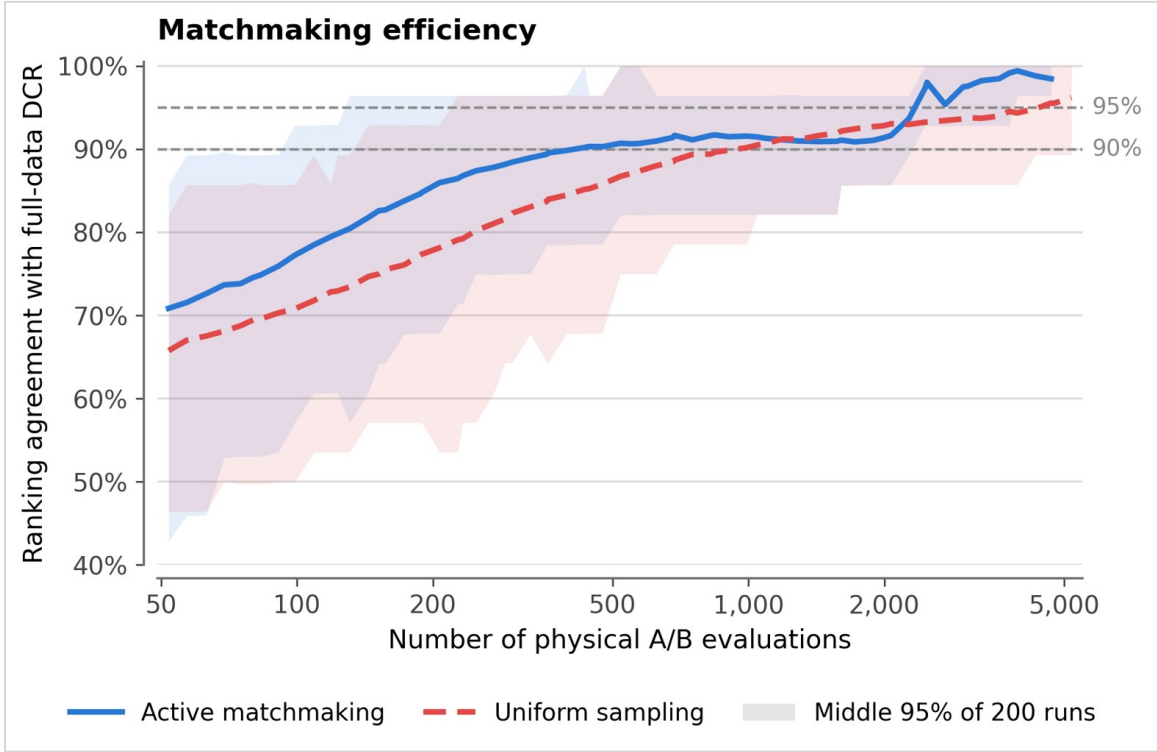
Agreement of rankings across four embodiments
(1 = perfect agreement)

- **π0.5:** 1st on DROID, YAM → 4th on G1
- **GR00T N1.7:** 1st on SO-101 → 4th on DROID, YAM
- **LingBot-VLA v2:** 5th on SO-101, YAM → 1st on G1
- **Note:** tasks and sites differ per embodiment, so gaps aren't due to morphology alone

Source: paper Figure 10a (Fine-tune Track mean PS ranks)

Reliable? Reproducible with enough data; active pairing halves trials





≈85%
DROID / YAM split-site agreement (ρ = 0.82)

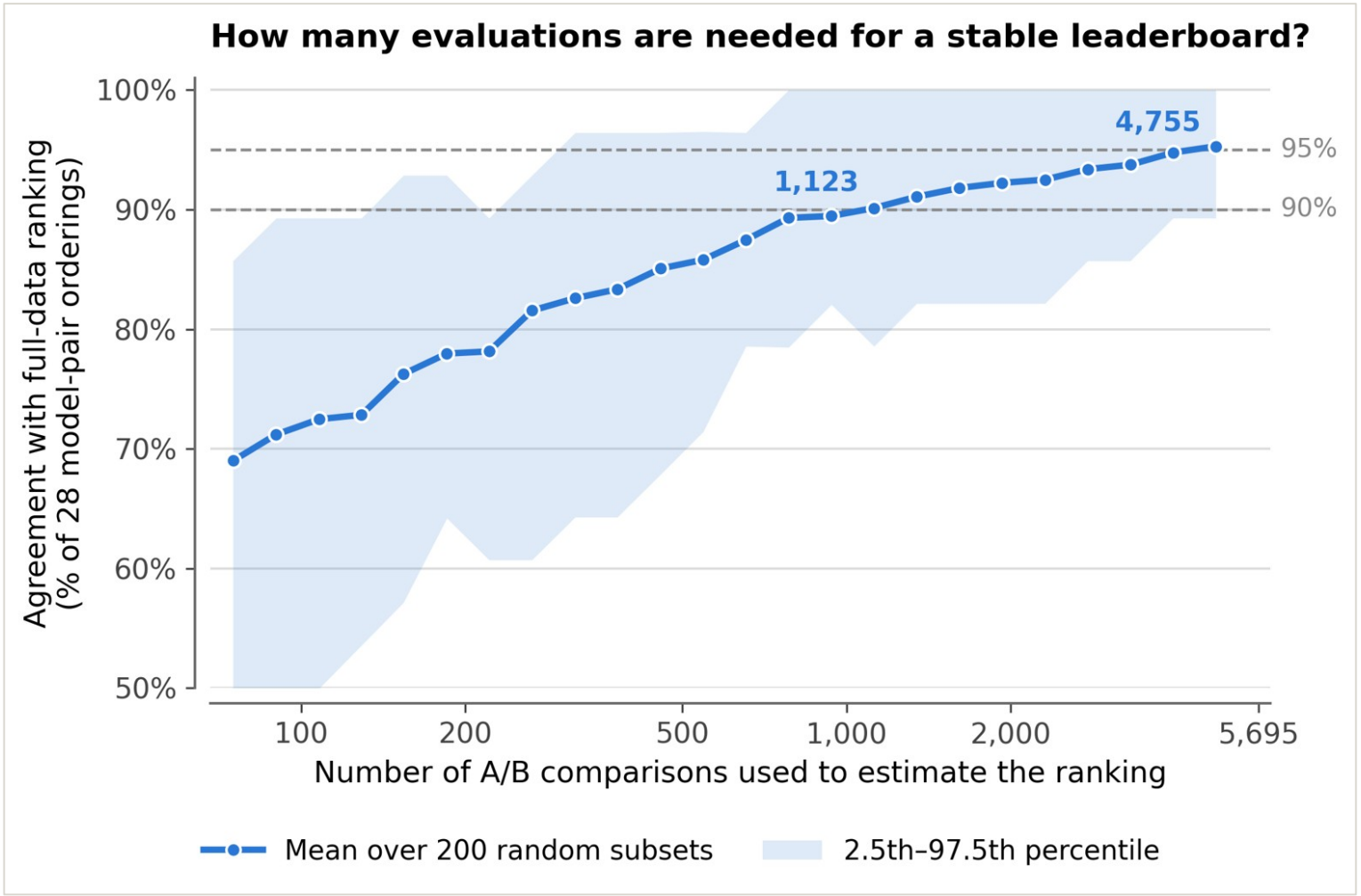
59%
SO-101 only ρ = 0.30; site-sensitive

400–500
Evals for active pairing to reach ~90% agreement

~1,000
Evals for uniform sampling to match it

Source: paper Figure 10b, Figure 11a (200 independent runs)

Mean ordering emerges early, but stability needs far more data



1,123 evals
Mean rank agreement reaches 90%

4,755 evals
Mean rank agreement reaches 95%

Wide quantile bands
Subsets of equal size can still yield clearly different rankings

Source: paper Figure 11b (200 random subsets per budget, 28 model pairs)

Assessment

Strengths & limits: valuable infrastructure; read rankings as snapshots

Contributions

- **Sustainable evaluation facility:** full-time operators, multinational multi-embodiment, unified serving; 10,000 hours of free academic evaluation pledged
- **Clear split between two tracks:** zero-shot generalization vs. fixed-data adaptation, with explicit controls
- **DCR ranking:** human-rated difficulty prior, keeps absolute success/failure; suits one trial per task
- **Informative findings:** agents are competitive, rankings don’ t transfer, humanoids need progress scores
- **Open resources:** 20 tasks × 100 demos, task cards, code and models

Limitations & open questions

- **Single-site Fine-tune Track:** one site and time window per task, not truly in the wild
- **Confounded training recipes:** GR00T froze its backbone only on SO-101, its only win
- **Uneven samples:** SO-101 only 496 runs; ~119 per agent on DROID
- **Cross-track comparison not clean:** checkpoints, tasks and metrics differ; only 5 models
- **Self-referential reliability:** full data partly collected via active pairing, somewhat circular

Four Takeaways

01 **Agents compete, but none succeed reliably**

Zero-shot frontier agents tie the best VLA; open-world success ⩽13% in every category, 0.4% at the hardest level

02 **No single leaderboard captures physical intelligence**

Cross-track $\rho \in [-0.40, +0.20]$, cross-embodiment $W = 0.24$; rankings flip often

03 **Progress scores are indispensable**

On G1, all 5 policies have 0 success and 13.3%–23.2% progress; failure points vary by model

04 **Large-scale real-robot eval is feasible and efficient**

Active pairing: ~90% agreement in 400–500 evals (uniform ~1,000); 95% stability needs ~4,755